

Lesson 3 · Quadratics and the Discriminant

NCEA Level 2 Mathematics · Algebra

By the end of today you can:

1. solve any quadratic: factorise, formula, complete the square
2. build the equation back from its roots
3. use $b^2 - 4ac$ to count roots and find an unknown k
4. prove a fact about the roots

Teaching line, not a question slide.

100 分钟。时间：对作业 0:00–0:05；Do Now 0:05–0:10；核心思想 0:10–0:12；Block 1 解方程 0:12–0:30；Block 2 由根求方程 0:30–0:42；Block 3 判别式 0:42–1:00；Block 4 证明 1:00–1:05；Classwork 1:05–1:25；真题演练 1:25–1:35；收尾 1:35–1:40。

发四份纸：Handout 3、Classwork 3、Practice set 3、Homework 3。

真题只有 4 页，时间够就让学生在 Practice set 上完整写，按评分表批。

Say it:

quadratic = kwod-RAT-ik

discriminant = dis-KRIM-ih-nunt

end of notes

Do Now

- 1 Evaluate $\log_2 64$.
- 2 Write $\log 12 - \log 3$ as a single log.
- 3 Solve $3^x = 20$.
- 4 Factorise $x^2 - 2x - 15$.
- 5 Expand $(2x - 3)^2$.

Q1 6

Q2 $\log 4$

Q3 $x = 2.727$

Q4 $(x - 5)(x + 3)$

Q5 $4x^2 - 12x + 9$

Short solution:

Q3 $x = \frac{\log 20}{\log 3}$

Q5 $(2x)^2 + 2(2x)(-3) + 9$

Watch for:

Q2 答 $\frac{\log 12}{\log 3}$: 减法对应分数在 log 里面 (L12)

Q5 漏掉中间项: $4x^2 + 9$

Diagnostic:

Q1–Q3 错: Lesson 2 没稳, 作业 2 的错题下节课再看一次

Q4、Q5 错: 今天 Block 1 前的 Quick review 要做足

end of notes

The one idea

$$(x - 3)(x + 2) = 0$$

$$(x - 3)(x + 2) = 6$$

A product is zero only when one factor is _____.

❓ Which equation can you solve by setting each bracket to zero? Why not the other?

Gap zero

Q the first; in the second, $x - 3 = 6$ does not follow

Short solution:

Q 第二个先展开: $x^2 - x - 12 = 0$, 再分解: $(x - 4)(x + 3) = 0$

Watch for:

学生写 $x - 3 = 6$: 乘积等于 6 时每个括号可以是任何数 (L19)

Ask:

$2 \times 3 = 6$, $1 \times 6 = 6$: 哪个括号等于 6? (不一定)

end of notes

Quick review: factorising

1 $x^2 + 5x + 6$

2 $x^2 - 9$

3 $2x^2 + 5x + 2$

4 $3x^2 - 6x$

Q1 $(x + 2)(x + 3)$

Q2 $(x - 3)(x + 3)$

Q3 $(2x + 1)(x + 2)$

Q4 $3x(x - 2)$

Short solution:

Q3 $ac = 4$: 4 和 1

Watch for:

Q4 写 $3(x^2 - 2x)$: 没有提完公因式 (L4)

end of notes

Set each factor to zero

$$(x - 3)(2x + 5) = 0$$

$$x - 3 = 0 \quad \text{or} \quad 2x + 5 = 0$$

$$x = \underline{\hspace{2cm}} \quad \text{or} \quad x = \underline{\hspace{2cm}}$$

❓ Why is the second answer not $x = 5$?

Gap 3; $-\frac{5}{2}$

Q $2x + 5 = 0$ gives $2x = -5$

Short solution:

Q 每个括号单独解：移项再除以系数

Watch for:

答 $x = -5$: 忘了除以 2

end of notes

Get = 0 first

Solve $x(x + 2) = 15$.

$$x^2 + 2x - 15 = 0$$

$$(x + 5)(x - 3) = 0$$

$$x = \underline{\hspace{2cm}} \quad \text{or} \quad x = \underline{\hspace{2cm}}$$

❓ Sam writes $x = 15$ or $x + 2 = 15$. What did Sam skip?

Gap $-5; 3$

Q rearranging to $= 0$ before setting brackets to zero

Short solution:

Q 展开，移项成 $= 0$ ，再分解；检查 $3 \times 5 = 15$

Watch for:

L19

Say it:

rearrange = ree-uh-RAYNJ

end of notes

When it won't factorise: the formula

$$x = \frac{-b \pm \sqrt{b^2 - 4ac}}{2a}$$

Solve $2x^2 - 5x - 1 = 0$: $a = 2$, $b = \underline{\hspace{2cm}}$, $c = -1$

$$x = \frac{-(-5) \pm \sqrt{(-5)^2 - 4(2)(-1)}}{2(2)} = \frac{5 \pm \sqrt{33}}{4}$$

❓ What is $-b$ when $b = -5$?

Gap -5

Q $+5$; $x = 2.686$ or $x = -0.186$

Short solution:

Q 把 a 、 b 、 c 连同符号放进括号再代入；公式在 formula sheet 上

Watch for:

写 -5 或 $-5^2 = -25$ ：没有括号 (L21)

只把 $\sqrt{33}$ 除以 4：整个分子都除以 $2a$

end of notes

Completing the square

Solve $x^2 - 6x + 2 = 0$ exactly.

$$x^2 - 6x = -2$$

$$(x - 3)^2 - \underline{\hspace{2cm}} = -2$$

$$(x - 3)^2 = 7$$

$$x = \underline{\hspace{2cm}}$$

❓ Where does the 3 in $(x - 3)$ come from?

Gap 9; $3 \pm \sqrt{7}$

Q half of the x coefficient, -6

Short solution:

Q $(x - 3)^2 = x^2 - 6x + 9$, 所以多出来的 9 要减掉
开方时要 $\pm$

Watch for:

只写 $3 + \sqrt{7}$: 漏掉负的那个

Exactly 表示留 surd, 不要小数

end of notes

A quadratic in disguise

Solve $x^4 - 13x^2 + 36 = 0$.

Let $u = x^2$:

$$u^2 - 13u + 36 = 0$$

$$(u - 4)(u - 9) = 0$$

$$x^2 = 4 \quad \text{or} \quad x^2 = 9$$

② How many values of x are there?

Q four: $x = \pm 2$ or $x = \pm 3$

Short solution:

Q 和 Lesson 2 的 3^{2x} 一样：认出“平方的平方”

Watch for:

只写 $u = 4$, $u = 9$ 就停：要换回 x

只写正根

end of notes

Spot the error (1)

Sam's answer has 2 errors. Find each one, fix it, and say why it loses marks.

Sam's answer

(a) Solve $x^2 + 3x = 10$. $x(x + 3) = 10$, so $x = 10$ or $x = 7$

(b) Solve $x^2 - 4x - 2 = 0$. $x = \frac{-4 \pm \sqrt{16 + 8}}{2} = -2 \pm \sqrt{6}$

1. (a) set to zero first: $x^2 + 3x - 10 = 0$, $(x + 5)(x - 2) = 0$, so $x = -5$ or $x = 2$ (L19)

2. (b) $b = -4$, so $-b = +4$: $x = 2 \pm \sqrt{6}$ (L21)

Short solution:

(a) $x = -5$ or $x = 2$

(b) $x = 4.449$ or $x = -0.449$

Watch for:

(a) Sam 的第二个答案是 $x + 3 = 10$ 解出来的 7: 两个错误都来自没等于 0

(b) 检查: $x = 2 + \sqrt{6}$ 代回去等于 0

end of notes

Practice

Q1 Solve $4x^2 = 11x + 3$.

Q2 Solve $x^2 + 4x - 7 = 0$. Give your answers to 3 decimal places.

Q3 Solve $x^4 - 10x^2 + 9 = 0$.

Q1 $x = 3$ or $x = -\frac{1}{4}$

Q2 $x = 1.317$ or $x = -5.317$

Q3 $x = \pm 1$ or $x = \pm 3$

Short solution:

Q1 $4x^2 - 11x - 3 = (4x + 1)(x - 3)$ (u)

Q2 $x = -2 \pm \sqrt{11}$ (u)

Q3 $(u - 1)(u - 9)$ (r)

Watch for:

Q1 先移项 (L19)

Q2 完全平方: $(x + 2)^2 = 11$ 也可以

end of notes

From the roots back to the equation

A quadratic has roots 4 and -3 .

$$x = 4 \Rightarrow (x - \underline{\quad}) \quad x = -3 \Rightarrow (x + \underline{\quad})$$

$$(x - 4)(x + 3) = 0$$

$$x^2 - x - 12 = 0$$

❓ Why is the bracket $(x - 4)$ and not $(x + 4)$?

Gap 4; 3

Q $x = 4$ must make the bracket zero: $4 - 4 = 0$

Short solution:

Q 根是让括号等于 0 的数；倒推回去符号相反

Watch for:

$(x + 4)(x - 3)$: 符号反了 (L17)

end of notes

Fraction roots: keep the coefficients whole

A quadratic has roots $\frac{3}{5}$ and -2 .

$$x = \frac{3}{5} \Rightarrow 5x = 3 \Rightarrow (\text{_____})$$

$$(5x - 3)(x + 2) = 0$$

$$5x^2 + 7x - 6 = 0$$

❓ Why use $(5x - 3)$ and not $(x - \frac{3}{5})$?

Gap $5x - 3$

Q it gives whole-number coefficients straight away

Short solution:

Q 用 $(x - \frac{3}{5})$ 也可以，但最后整个方程要乘 5

Watch for:

题目要 whole numbers 时留分数：只算 u (L18)

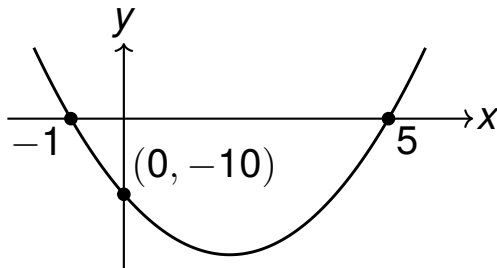
end of notes

Equation of a parabola from its graph

The parabola crosses the x -axis at -1 and 5 , and passes through $(0, -10)$.

$$y = a(x + 1)(x - 5)$$

$$-10 = a(\underline{\hspace{1cm}})(\underline{\hspace{1cm}})$$



❓ What does the point $(0, -10)$ give you?

Gap 1; -5

Q the value of a : $a = 2$, so $y = 2x^2 - 8x - 10$

Short solution:

Q 截距给出两个括号，第三个点求 a ；代 $x = 0$, $y = -10$

Watch for:

忘了 a ，写 $y = (x + 1)(x - 5)$ ：这条抛物线过 $(0, -5)$ ，不过 $(0, -10)$

Say it:

parabola = puh-RAB-uh-luh

end of notes

Spot the error (2)

Find a quadratic with roots $\frac{2}{3}$ and -4 , with whole-number coefficients.

Sam's answer has 2 errors. Find each one, fix it, and say why it loses marks.

Sam's answer

$$\left(x + \frac{2}{3}\right)(x - 4) = 0 \qquad x^2 - \frac{10}{3}x - \frac{8}{3} = 0$$

1. the root $\frac{2}{3}$ gives $(3x - 2)$, and the root -4 gives $(x + 4)$: both signs reversed (L17)

2. the coefficients are not whole numbers (L18)

Short solution:

$$(3x - 2)(x + 4) = 3x^2 + 10x - 8 = 0$$

Watch for:

检查: $x = \frac{2}{3}$ 代进 Sam 的括号 $\left(x + \frac{2}{3}\right)$ 不等于 0

end of notes

Practice

Q1 Find a quadratic with roots -5 and $\frac{1}{2}$, with whole-number coefficients.

Q2 Find a quadratic with roots $\frac{3}{4}$ and $-\frac{2}{5}$, with whole-number coefficients.

Q3 A parabola has x -intercepts 2 and 6 and y -intercept 24. Find its equation.

Q1 $2x^2 + 9x - 5 = 0$

Q2 $20x^2 - 7x - 6 = 0$

Q3 $y = 2x^2 - 16x + 24$

Short solution:

Q1 $(x + 5)(2x - 1)$ (u)

Q2 $(4x - 3)(5x + 2)$ (r)

Q3 $y = a(x - 2)(x - 6)$, $24 = 12a$, $a = 2$ (r)

Watch for:

Q2 $-\frac{2}{5}$ 给 $(5x + 2)$ (L17)

end of notes

Quick review: inequalities

- 1 Solve $3x - 2 > 7$.
- 2 Solve $12 - 4k < 0$.
- 3 Solve $-2p \geq 8$.

Q1 $x > 3$

Q2 $k > 3$

Q3 $p \leq -4$

Short solution:

Q2 $-4k < -12$, 除以 -4 要变号

Watch for:

Q2、Q3 不变号：判别式求 k 的范围时同样的错

Say it:

inequality = in-ih-KWOL-ih-tee

end of notes

What $b^2 - 4ac$ tells you

Gap two; one (repeated); no real

Q under the square root: a negative number has no real square root

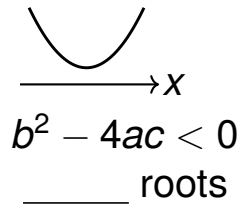
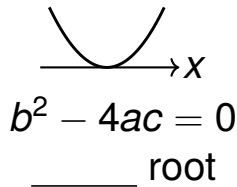
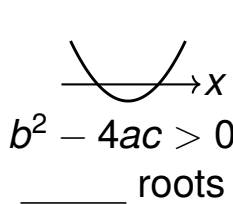
Short solution:

$Q \pm \sqrt{\Delta}$: 正数给两个不同的根, 0 给一个, 负数没有实根

Watch for:

“Equal roots”、“exactly one solution”、“touches the x-axis” 都是 $= 0$

end of notes



② Where in the formula does $b^2 - 4ac$ sit, and why does its sign matter?

Counting roots

How many real roots has $3x^2 - 4x + 2 = 0$?

$$\begin{aligned}b^2 - 4ac &= (-4)^2 - 4(3)(2) \\ &= 16 - 24 = ____\end{aligned}$$

❓ What does a negative discriminant tell you about the graph?

Gap -8

Q it never touches the x-axis: no real roots

Short solution:

Q $\Delta < 0$, 所以没有实根; 不用解方程

Watch for:

$-4^2 = -16$: b 没加括号 (L21)

end of notes

Finding k : equal roots

$x^2 + kx + 16 = 0$ has equal roots. Find k .

$$k^2 - 4(1)(16) = \underline{\hspace{2cm}}$$

$$k^2 = 64$$

❓ Why are there two values of k ?

Gap 0

Q $k^2 = 64$ gives $k = 8$ or $k = -8$

Short solution:

Q equal roots 就是 $\Delta = 0$; 平方根有正负

检查 $k = 8$: $x^2 + 8x + 16 = (x + 4)^2$

Watch for:

只给 $k = 8$

end of notes

Finding k : an inequality

$2x^2 + 6x + p = 0$ has two different real roots. Find the values of p .

$$6^2 - 4(2)(p) > 0$$

$$36 - 8p > 0$$

$$-8p > -36$$

② When you divide by -8 , what happens to the $>$ sign?

Q it turns round: $p < 4.5$

Short solution:

Q two different real roots 是 > 0 ; 除以负数变号

Watch for:

答 $p > 4.5$ (不变号)

用 $\geq$: “different” 不能等于 (L20)

end of notes

When k is in more than one place

$kx^2 + 4x + (k - 3) = 0$ has equal roots. Find k .

$$a = k, \quad b = 4, \quad c = \underline{\hspace{2cm}}$$

$$4^2 - 4(k)(k - 3) = 0$$

$$k^2 - 3k - 4 = 0$$

② What are the two values of k ?

Gap $k - 3$

Q $k = 4$ or $k = -1$

Short solution:

Q $16 - 4k^2 + 12k = 0$, 除以 -4 : $k^2 - 3k - 4 = (k - 4)(k + 1)$ (r)

Watch for:

c 不加括号: $4k \times k - 3$ (L21)

这是 Lesson 1 的展开 + 今天的分解

end of notes

Spot the error (3)

$x^2 - 5x + k = 0$ has no real roots. Find the values of k .

Sam's answer has 2 errors. Find each one, fix it, and say why it loses marks.

Sam's answer

$$b^2 - 4ac = -5^2 - 4(1)(k) = -25 - 4k$$

no real roots, so $-25 - 4k > 0$, so $k < -6.25$

1. $(-5)^2 = 25$, not -25 : bracket the negative b (L21)

2. no real roots needs $b^2 - 4ac < 0$, not > 0 (L20)

Short solution:

$$25 - 4k < 0, \quad -4k < -25, \quad k > 6.25$$

Watch for:

检查 $k = 7$: $\Delta = 25 - 28 < 0$, 没有实根

end of notes

Practice

Q1 Find the discriminant of $4x^2 + 12x + 9 = 0$. What does it tell you?

Q2 $x^2 - 6x + k = 0$ has no real roots. Find the values of k .

Q3 $mx^2 + 2mx + 3 = 0$ has equal roots, and $m \neq 0$. Find m .

Q1 0: one repeated root, $x = -\frac{3}{2}$

Q2 $k > 9$

Q3 $m = 3$

Short solution:

Q1 $144 - 144 = 0$; $(2x + 3)^2 = 0$ (u)

Q2 $36 - 4k < 0$ (r)

Q3 $4m^2 - 12m = 0$, $4m(m - 3) = 0$, $m \neq 0$ (r)

Watch for:

Q3 给出 $m = 0$: 题目说 $m \neq 0$, 而且 $m = 0$ 时不是二次方程

end of notes

Proving a fact about the roots

Show that one solution of $x^2 - 5kx + 4k^2 = 0$ ($k \neq 0$) is four times the other.

$$(x - k)(x - \underline{\hspace{2cm}}) = 0$$

$$x = k \quad \text{or} \quad x = 4k$$

❓ What sentence must you write at the end?

Gap 4k

Q “so one solution is four times the other”

Short solution:

Q 两个数乘积 $4k^2$ 、和 $-5k$: $-k$ 和 $-4k$

也可以用公式: $\Delta = 9k^2$, $x = \frac{5k \pm 3k}{2}$

Watch for:

只写 $x = k$ 或 $4k$ 没有结论: 评分表要 correct conclusion

Ask:

为什么题目说 $k \neq 0$? ($k = 0$ 时两个解都是 0)

end of notes

Practice

Q1 Show that one solution of $2x^2 - 7kx + 3k^2 = 0$ ($k \neq 0$) is six times the other.

Q2 Show that $x^2 + (k + 2)x + 2k = 0$ has real roots for every value of k .

Q1 $x = \frac{k}{2}$ or $x = 3k$, and $3k = 6 \times \frac{k}{2}$

Q2 $\Delta = (k - 2)^2 \geq 0$ for every k

Short solution:

Q1 $(2x - k)(x - 3k) = 0$ (r)

Q2 $(k + 2)^2 - 8k = k^2 - 4k + 4 = (k - 2)^2$; 平方不会是负数 (t)

Watch for:

Q2 只代几个 k 试: 不是 every value (L24 下节课会讲)

end of notes

Classwork 3 · about 20 minutes

- **Part A:** solving, the discriminant, roots to an equation
- **Part B:** finding k and p ; a parabola from three facts
- **Part C:** real roots for every k , explained
- **Extension**, if you finish

Teaching line, not a question slide.

答案在 classwork3_answers.pdf (同版面, 红色答案)。

A1 $\frac{1}{2}$ 、 -4 ; A2 7 、 -4 ; A3 0.869 、 -1.535 ; A4 -12 , 没有实根; A5 $3x^2 + x - 2 = 0$

B1 $k = \pm 12$; B2 $p < 16$; B3 $y = 2x^2 + 2x - 12$

C: $\Delta = (k - 3)^2 \geq 0$; $k = 3$ 时相等

Extension: $x = \pm 1$ 或 ± 2

end of notes

(b) A quadratic equation has solutions of $x = -\frac{2}{3}$ and $x = 4$.

Find the original equation, giving your answer in the form of $ax^2 + bx + c = 0$, where a , b , and c are whole numbers.

(b) $3x^2 - 10x - 8 = 0$

Short solution:

(b) $(3x + 2)(x - 4) = 0$

评分：系数不是整数、或因式形式错但展开一致，只算 u；正确方程是 r

Watch for:

$-\frac{2}{3}$ 给 $(3x + 2)$ (L17)

end of notes

(c) Consider a quadratic equation in the form $x^2 - 3kx + 2k^2 = 0$, where k is a non-zero constant.

Show that one solution is twice the other solution.

(c) $(x - k)(x - 2k) = 0$, so the solutions are k and $2k$: one is twice the other

Short solution:

(c) 或者: $\Delta = 9k^2 - 8k^2 = k^2$, $x = \frac{3k \pm k}{2}$

评分: 分解出方程或算出判别式是 u; 有效过程加结论是 r

Watch for:

最后一句结论不能省

end of notes

(a) A quadratic equation, $ax^2 + bx + c = 0$, has solutions of $\frac{1}{3}$ and $\frac{-2}{7}$.

Find the values of the integers a , b , and c .

(a) $a = 21, b = -1, c = -2$

Short solution:

(a) $(3x - 1)(7x + 2) = 21x^2 - x - 2$; 评分: correct values of a, b, c (u)

Watch for:

$a = -21, b = 1, c = 2$ 也对 (两边乘 -1)

$\frac{-2}{7}$ 给 $(7x + 2)$ (L17)

end of notes

(b) (i) What is the discriminant of the equation $2x^2 - 12x + 7 = 0$?

(ii) Suppose $y = 2x^2 - 12x + k$, where k is a constant.

For what value of k will the equation $y = 0$ have exactly one solution?

(i) 88

(ii) $k = 18$

Short solution:

(i) $(-12)^2 - 4(2)(7) = 144 - 56$ (u)

(ii) $(-12)^2 - 4(2)(k) = 0$, $8k = 144$: 代入是 u, 正确的 k 是 r

Watch for:

exactly one solution 是 $= 0$ (L20)

end of notes

Ways to lose marks

- **L17** Root $\frac{2}{3}$ gives $(3x - 2)$; root -4 gives $(x + 4)$
- **L18** Whole-number coefficients when the question asks
- **L19** Rearrange to $= 0$ before setting brackets to zero
- **L20** Two different: > 0 ; equal: $= 0$; none: < 0
- **L21** $(-5)^2 = 25$: brackets round b and k

Teaching line, not a question slide.

讲义上有 L17–L21 的完整说明

让学生对照 Classwork 和 Practice set 的错，说出是哪一个 L

end of notes

Summary

- Get $= 0$ first; then factorise, or use the formula with brackets.
- Roots to equation: root $\frac{p}{q}$ gives the factor $(qx - p)$.
- $b^2 - 4ac$: > 0 two, $= 0$ one, < 0 none.
- Proofs end with the sentence the question asks for.

Teaching line, not a question slide.

让学生说出三种判别式情况对应的图

end of notes

Homework 3 · about 60 minutes

- **Quick review:** four short items
- **Question One:** solving quadratics
- **Question Two:** equations from roots and graphs
- **Question Three:** the discriminant

Each question is marked like the real paper: N0 to E8.

Teaching line, not a question slide.

答案在 hw3_answers.pdf; 每题的 (e) 是 Excellence 题
下节课开头 5 分钟对答案

end of notes